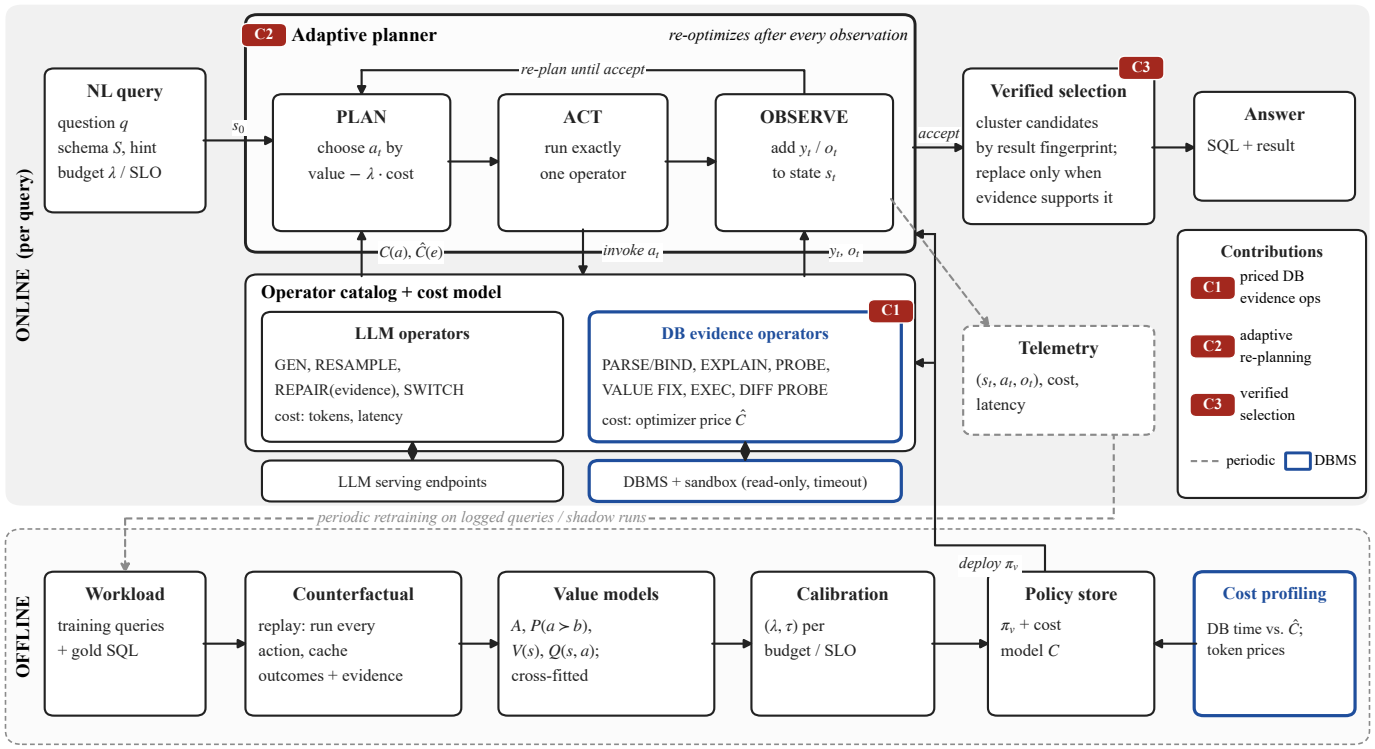

ONLINE (per query)
C2 Adaptive planner
re-optimizes after every observation
re-plan until accept
NL query
question q
schema S, hint
budget λ / SLO
s_0
PLAN
choose a_t by
value − $\lambda \cdot$ cost
ACT
run exactly
one operator
OBSERVE
add y_t / o_t
to state s_t
accept
C3
Verified selection
cluster candidates
by result fingerprint;
replace only when
evidence supports it
Answer
SQL + result
$C(a)$, $\hat{C}(e)$
invoke a_t
y_t, o_t
Operator catalog + cost model
LLM operators
GEN, RESAMPLE,
REPAIR(evidence), SWITCH
cost: tokens, latency
C1
DB evidence operators
PARSE/BIND, EXPLAIN, PROBE,
VALUE FIX, EXEC, DIFF PROBE
cost: optimizer price $\hat{C}$
LLM serving endpoints
DBMS + sandbox (read-only, timeout)
Telemetry
(s_t, a_t, o_t), cost,
latency
Contributions
C1 priced DB evidence ops
C2 adaptive re-planning
C3 verified selection
periodic
DBMS
periodic retraining on logged queries / shadow runs
OFFLINE
Workload
training queries
+ gold SQL
Counterfactual
replay: run every
action, cache
outcomes + evidence
Value models
A, $P(a \succ b)$,
$V(s)$, $Q(s, a)$;
cross-fitted
Calibration
(λ, τ) per
budget / SLO
deploy π_v
Policy store
π_v + cost
model C
Cost profiling
DB time vs. $\hat{C}$;
token prices